Fiscal Year:	FY 2005	**Task Last Updated:**	FY 06/22/2005
PI Name:	Zerwekh, Joseph E Ph.D.		
Project Title:	Prevention of Microgravity-Induced Stone Risk by KMgCitrate		
Division Name:	Human Research		
Program/Discipline:	NSBRI Teams		
Program/Discipline--Element/Subdiscipline:	NSBRI Teams--Bone Loss Team		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
PI Email:	Joseph.Zerwekh@utsouthwestern.edu	**Fax:**	FY 214-648-2526
PI Organization Type:	UNIVERSITY	**Phone:**	214-648-3872
Organization Name:	The University of Texas Southwestern Medical Center at Dallas		
PI Address 1:	5323 Harry Hines Blvd		
PI Address 2:			
PI Web Page:			
City:	Dallas	**State:**	TX
Zip Code:	75390-8885	**Congressional District:**	30
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	NSBRI
Start Date:	06/01/2001	**End Date:**	09/30/2005
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Antich, Peter (The University of Texas Southwestern Medical Center at Dallas) Pak, Charles (The University of Texas Southwestern Medical Center at Dallas)		
Grant/Contract No.:	NCC 9-58-BL00210		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The hypothesis to be tested in this project is that potassium magnesium citrate supplementation will attenuate the increased risk for stone formation and diminish microgravity-induced bone loss. This hypothesis will be tested during five weeks of bed rest in normal volunteers through three specific aims: Specific aim 1: Assess the efficacy of supplementation with potassium magnesium citrate (KMgCit) in preventing microgravity-induced increased risk of stone formation. The hypothesis to be tested is that supplementation with KMgCit will avert the increase in urinary stone forming risk during bed rest by increasing urinary pH, promoting magnesium complexation of oxalate, and raising urinary citrate, an established inhibitor of crystal agglomeration. Specific aim 2: Evaluate the effect of KMgCit supplementation in averting diminished muscle magnesium and potassium concentrations that may occur during microgravity-induced muscle atrophy. The hypothesis to be tested is that by providing adequate potassium and magnesium supplementation, the loss of these elements from muscle during prolonged bed rest will be averted. Specific aim 3: Assess the efficacy of supplementation with KMgCit in reducing microgravity-induced increases in bone resorption and urinary calcium. The hypothesis to be tested is that during five weeks of bed rest where bone loss is known to occur, supplementation with KMgCit will prevent the negative impact of low alkali intake and mild acidosis on bone resorption, urinary calcium loss, and further reduce the increased risk for calcium oxalate stone formation. Although the study design is a placebo-controlled randomized double-blind trial, we performed an interim analysis utilizing the laboratory findings from 15 subjects who have successfully completed the study. The study blind was maintained and the analysis was directed specifically at the hypothesis addressed in specific aim 1. We analyzed the data by comparing the differences between baseline and the last week of bed rest (week 5) for selected urinary analytes for placebo- versus KMgCit-treated subjects. Significance between treatment groups was assessed from the Satterthwaite t-test assuming unequal variances. There were 7 subjects in the placebo group (4 men, 3 women, mean age 40 ± 8 years) and 8 subjects in the KMgCit group (5 men, 3 women, mean age 37 ± 10 years). As expected, KMgCit significantly increased urinary pH by 0.75 pH units as compared to the 0.13 unit change in the placebo group (p=0.018). Urinary calcium and phosphorus increased comparably in both groups during bed rest. Urinary oxalate declined in both groups with no significant difference between groups. For placebo-treated subjects, urinary citrate declined in 4 (mean decline 142 ± 109 mg/d) and increased in 3 (mean increase 159 ± 92 mg/d). On the other hand, urinary citrate increased in all subjects receiving KMgCit by 652 ± 457 mg/d (p=0.015 vs placebo). Net GI absorption of alkali was significantly increased for KMgCit-treated subjects (p=0.0003). There were also significant reductions in urinary ammonium and total titratable acidity with KMgCit treatment, indicative of an overall reduction in net acid excretion. Despite the increase in urinary calcium in both groups, the relative saturation ratio (RSR) for calcium oxalate declined significantly in the KMgCit-treated patients compared to placebo (p=0.034). There was no overall change in either the RSR or formation product ratio (FPR) for brushite in this analysis. There were also no significant differences in uric acid excretion or in the RSR for sodium urate. On the other hand, the concentration of undissociated uric acid in urine declined more during KMgCit treatment but the difference between placebo and KMgCit-treated subjects did not quite attain significance at this analysis (p=0.069). This interim analysis is consistent with the hypothesis that provision of alkali as KMgCit attenuates the increased risk of stone formation associated with skeletal unloading through an increase in urinary pH and reduced ionic calcium by citrate binding. The fact that the relative saturation ratio for calcium oxalate fell significantly despite an overall increase in total urinary calcium attests to this. At present, there is no suggestion that provision of magnesium reduced intestinal oxalate absorption. It is hoped that with the addition of the last 5 subjects there will emerge additional significant beneficial changes in urine composition to reduce renal stone risk. Nevertheless, these preliminary findings support the hypothesis that this “nutriceutical” countermeasure appears to reduce the risk of renal stone formation by significantly reducing urinary saturation of calcium oxalate and uric acid, two principle stone-forming salts. We anticipate that we will complete all subject evaluations by early Spring 2005. We will commence with potassium and magnesium analyses in the muscle biopsies to determine whether KMgCitrate may avert loss of these ions from muscle during unloading (specific aim 2). In addition, we anticipate completing all of the assays of the biochemical markers of bone turnover. This data will be combined with changes in bone density and in bone quality as assessed by in vivo critical angle ultrasound reflectometry. These analyses should provide a clear indication as to whether this countermeasure may also have efficacy in reducing the loss of bone mineral invariably seen during skeletal unloading (specific aim 3).
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Nephrolithiasis is a significant medical condition that was responsible for 1.3 million physician office visits in 1997 and an estimated 290,000 hospital discharges in 1999 due to a stone-related problem. It is apparent that any advances made in our understanding of the metabolic, nutritional, and environmental factors that contribute to increased stone risk in space will have direct application to this pervasive problem on Earth. Although there may be differences in the etiology of hypercalciuria between astronauts and Earth-based stone formers, there are striking similarities in the changes that can occur in the urinary environment predisposing to stones. Nutritional, as well as pharmacologic based countermeasures, that may prove to be effective in reducing stone risk in astronauts, would be a welcomed additional tool for the management of Earth-based disease. In addition, if our study should demonstrate a beneficial effect of our countermeasure in reducing bone mineral loss, it could have far-reaching implications for the 25 million Americans who are osteopenic or osteoporotic.
Task Progress:	We have nearly completed our proposed bed rest study. Due to restrictions in bed utilization by the GCRC from 2 beds for this study to one bed, we have had to extend our timeline by 1 year for completion. It is anticipated that the 19th subject will complete the study by year’s end and that the last subject will be completed by early Spring of 2005.
Bibliography Type:	Description: (Last Updated: 02/05/2007)
Articles in Peer-reviewed Journals	Zerwekh, J. E. "Nutrition and renal stone disease in space Nutrition" 857-863 , Oct-2002